

Docker Networks :

a Docker network is a communication channel that allows containers to talk to each other, the host, or the outside world.

1. Bridge Network

- Definition : A private internal network created by docker on the host, containers get their own IPs and can talk to each other via the network.
- Use : Good for connecting multiple containers on the same host while keeping them isolated from the outside unless explicitly exposed.

Example:-

Create custom bridge network

- `docker network create mybridge`

Run two containers in this network

- `docker run -dit --name c1 --network mybridge alpine sh`
- `docker run -dit --name c2 --network mybridge alpine sh`

From c1, you can ping c2 by container name

- `docker exec -it c1 ping c2`

2. Host Network.

• Definition : The container shares the host's network stack, it uses the host's IP address directly, no virtual interface

- Use : Best for high-performance networking and when the container needs to use host network ports without NAT overhead

Example

- `docker run -dit --name web --network host nginx`
Now nginx runs on host's IP and port 80 directly.

3. None Network.

- Definition :- The container has no network interface except the loopback interface (lo).
- Use : For fully isolated workloads that don't need any network access (like a background data processor).

Example :-

- `docker run -dit --name isolated --network none alpine sh`

inside this container, no internet or other container communication.

Useful Commands.

<u>Command</u>	<u>Purpose</u>
• <code>docker network ls</code>	List all Docker networks
• <code>docker network inspect <network></code>	show details of a network
• <code>docker network rm <network></code>	Delete a network
• <code>docker network create mynet --driver=bridge</code>	Create a custom bridge network
• <code>ifconfig / ip a</code>	show network interfaces
• <code>curl <url></code>	